

LLM Judge Validation: Cross-Model Evaluation (Balanced)

v5 balanced: GPT-4o-mini extractions vs human-verified GT

Partial detection: 27/32 (84%) | Sample: Correct=17, Partial=8, Incorrect=15

Table 1: Per-Judge Performance

Judge	Overall	Correct Acc	Partial Acc	Incorrect Acc	Cohen's κ
GPT-4o-mini	24/39 (62%)	59% (10/17)	88% (7/8)	50% (7/14)	0.433
GPT-4o	27/40 (68%)	65% (11/17)	88% (7/8)	60% (9/15)	0.508
GPT-5.4	26/40 (65%)	71% (12/17)	75% (6/8)	53% (8/15)	0.465
Claude 4.5	27/40 (68%)	65% (11/17)	88% (7/8)	60% (9/15)	0.508

Table 2: Comparison Across Experiments

Metric	v1-v4 (Circular)	v5 Balanced
Partial detection	0/24 (0%)	27/32 (84%)
Overall agreement	84-88%	65%
Cohen's κ (best)	0.33-0.47	0.508
Sample balance	41/6/3 (skewed)	17/8/15

v1-v4 tested Claude extractions against Claude-derived GT (circular — identical text). v5 uses GPT-4o-mini extractions that genuinely differ from human-verified GT, enabling proper evaluation. The balanced sample (17/8/15) ensures reliable per-category accuracy. All judges detect Partially Correct cases 75-88% of the time when extraction text truly differs.